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DEPARTMENT OF CHEMICAL ENGINEERING

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AquaCaptor - Water Harvesting with MOF/Salt Composite

GROUP 10

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1. Executive Summary

Globally, there is a growing need for and difficulty in accessing clean water, and solutions are being proposed to overcome this challenge. AquaCaptor is a water harvesting system developed as a solution to the water crisis, integrated into home air conditioners. The project is an innovative add-on technology that brings clean water into the home instead of obtaining it from large, complex systems.

The AquaCaptor project differs from existing atmospheric water harvesting technologies in its material composition. The developed MOF/Salt composite incorporates dual entrapment technology, which is unique in this respect. This dual entrapment technology minimizes the leakage of CaCl_2 salt, which both reduces moisture retention capacity and damages the integrated air conditioner. CaCl_2 salts are trapped in the pores of the UiO-66 MOF and re-coated using Chitosan/Alginate biopolymers, thus preventing the risk of leakage. Furthermore, the project utilizes the waste heat of the integrated air conditioner to desorb the trapped moisture without requiring an additional energy source.

Market Analysis and Customer Needs

The project adopts a B2B (business-to-business) model, aiming to establish partnerships with HVAC (Heating, Ventilating, and Air Conditioning) manufacturers and sell prototypes to these companies, rather than selling directly to end consumers. Target Customers: The "room air conditioner" segment, with a global usage rate of 42% (See **Figure 3.1-2**), is a primary target. Target customers include both domestic (Arçelik, Vestel, etc.) and global companies (Samsung, Daikin, etc.).

Both global and domestic target customers are driven by the need to design products in compliance with environmental regulations, differentiate their portfolios, and achieve a superior market position. From the end-user perspective, technology addresses the critical demand for easy access to clean water in the face of increasing water scarcity. Consequently, the project aims to contribute to and capitalize on this expanding market, which is projected to experience a compound annual growth rate of 10.4% (See **Figure 3.1-1**).

Financial Highlights

The total budget required for completing the R&D processes of the composite to be developed and proposing the prototype has been determined as USD 56,436. The financial performance of the project is as follows:

- It is aimed at achieving an annual sales revenue of USD 211,267 and a net profit of USD 47,393 by the end of the 4th year of the project.
- The IRR (Internal Rate of Return) of the project is calculated as 39.37% and the CoC as 21.65%.
- According to the cumulative cash flow, the project will become profitable at the end of the 4th year.

Project Schedule, Sustainability and Organization

- The project is designed to start on October 23, 2025, and be completed on December 31, 2028. The project duration is 38 months. 2026 is defined as the analysis and material selection year, 2027 as the product development year, and finally 2028 as the testing process and launch year.
- The developed project is in line with UN Sustainable Development Goals such as clean energy, clean water and responsible production.
- The R&D studies will be conducted at Hacettepe University Department of Chemical Engineering, and prototype studies will be carried out in collaboration with Hacettepe Technopark. Analysis processes will be supported by external laboratories such as METU MERLAB or Bilkent UNAM.
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** The project's managerial scope, financial details, and risk analyses are shown in Appendix A: Project Chart at the end of the report, while the commercialization methods and value proposition of the project are shown in **Appendix B: Business Canvas Model**.*

2. Goals and Scope

2.1. Project Goals

The primary goal of the AquaCaptor project is to develop a sustainable, energy-efficient atmospheric water harvesting (AWH) system that integrates seamlessly with existing HVAC infrastructure. Our project aims to address global water scarcity by transforming waste heat from air conditioners into a resource for clean water production using innovative MOF/Salt composite materials. This system is specifically designed for cities and coastal regions with high atmospheric humidity levels, aiming to maximize water production efficiency by leveraging the high moisture potential in these areas.

At the core of the project is the development of a MOF/salt composite material that can rely on adsorbing and releasing the water under realistic operating conditions. The project aims to synthesize and optimize a chitosan/alginate-coated $\text{CaCl}_2@\text{UiO-66}$ composite that combines high water uptake with mechanical stability and long service life as planned. Special attention is given to preventing salt leakage, maintaining performance over adsorption–desorption cycles repeating, and ensuring that the material can resist the temperature range typically found in air conditioning outdoor units.

Another important goal of the project is to move beyond material development and demonstrate a working system. For this reason, AquaCaptor aims to design and build a laboratory-scale functional prototype that can be integrated with an air conditioning unit and tested under controlled but realistic conditions. The prototype is intended to prove that the concept works as a complete system, producing water at meaningful rates while meeting basic quality and safety requirements.

The project goals are categorized into functional, business, technological, and quality objectives, prioritized to guide the execution phase effectively.

Table 2.1-1. Project goals and their priorities

Project Goal	Priority	Comment/Description/Reference
Functional Goals:	1	Core Functionality
<i>Water Generation from Waste Heat</i>	1	The system must effectively desorb moisture using only the waste heat (50–60°C) from the AC unit, without additional electrical input.
<i>HVAC Integration (Retrofit)</i>	1	The prototype must be designed as a modular "Add-on Kit" compatible with standard residential AC outdoor units.
Technological Goals:	1	Innovation & Feasibility
<i>Green Synthesis of UiO-66</i>	1	Synthesis of the Metal-Organic Framework using water-based/green solvents (formic acid instead of DMF) to ensure environmental safety.
<i>Dual Confinement Strategy</i>	1	Successful coating of CaCl ₂ impregnated UiO-66 with Chitosan/Alginate to prevent salt leakage and corrosion.
Quality Goals:	2	Compliance & Safety
<i>Water Potability</i>	1	The harvested water must meet WHO drinking water standards (TDS < 500 ppm, pH 6.5-8.5).
<i>Zero Leakage & Corrosion</i>	2	The composite material must show 0% salt leakage to prevent damage to the HVAC heat exchangers.
Business Goals:	2	Viability
<i>Cost-Effectiveness</i>	2	Keeping the prototype production cost within the budget limits and achieving a projected IRR of 27.96%.
<i>Intellectual Property (IP)</i>	2	Securing the formulation of the hybrid composite for future licensing opportunities (B2B Model).
Constraints:	-	Limitations
<i>Environmental Conditions</i>	-	The system performance is dependent on relative humidity (optimized for >40% RH).
<i>Space Limitations</i>	-	The design must be compact enough not to obstruct the airflow of the AC unit.

2.2. Project Scope

The scope of the AquaCaptor project covers the research, development, and validation steps required to turn an atmospheric water harvesting concept into a proven prototype. This includes laboratory-scale synthesis of the UiO-66 MOF, salt loading to enhance moisture capture, and polymer coating to improve mechanical strength and prevent material degradation. Material characterization and performance testing are important and required parts of the scope, ensuring that each development step is supported by experimental data.

The project scope also includes the design and assembly of a functional prototype that can be mounted onto an air conditioning system. Mechanical design, cartridge fabrication using 3D

printing, and basic heat transfer integration are carried out to create a system that reflects real operating conditions. The prototype is tested for water production capacity, adsorption–desorption behavior, thermal stability, durability, and water quality.

Beyond the technical work, the scope extends to project management, market analysis, cost evaluation, and risk assessment activities that are necessary to guide development and support future decision making. Environmental and safety aspects, such as laboratory waste handling, regulatory considerations, and sustainability impacts, are addressed throughout the project.

The project does not include mass production since the project is weighed on prototype productions that are able to be specialized within stakeholder demands, factory setup, or direct consumer sales. AquaCaptor is intentionally positioned as an R&D and pre-commercial initiative. All work is carried out using existing university laboratories, Technopark facilities, and external testing services, in line with an asset-light development approach and a B2B licensing strategy.

3. Execution

3.1. Marketing & Sales

3.1.1. Marketing Plan

The AquaCaptor Project's marketing strategy offers a solution to one of today's biggest problems, the water crisis, through green methods, and the goal of using energy efficiently.

Marketing and Advertising Goals

The company's primary marketing objective is to position AquaCaptor in the HVAC market as an environmentally friendly and innovative component that produces clean and usable water. The company's advertising strategy is as follows:

- ***Securing B2B Partnerships:*** Instead of selling the product directly to the end consumer, the company aims to establish strategic partnerships with HVAC manufacturers such as Arçelik, Vestel, and Mitsubishi Electric, both in Turkey and globally, to integrate AquaCaptor into their products.
- ***Sustainability Perspective:*** By emphasizing that the product is produced using "Green Synthesis" (DMF-free), unlike traditional methods, and that it recycles waste heat to produce water, the company aims to highlight its environmental awareness and build brand value in the market.
- ***Dual-Benefit Emphasis:*** By positioning AquaCaptor as a sustainable water solution provider in the HVAC market, the company aims to emphasize that the end product used by the consumer is not just a heating or cooling mechanism, but an active water producer.

Marketing Position

AquaCaptor is a startup company currently in the R&D and prototyping phase. The project's current market position is a pre-commercialization stage, focusing on pilot production processes. Unlike its competitors, it offers a solution that is both environmentally friendly and consumer-friendly by following a strategy of offering a product that can be integrated into existing infrastructure.

Timeline for Market Strategy and Implementation

The marketing strategy has been planned in accordance with the WBS. **Table 3.1** shows the completion schedule of marketing activities.

Table 3.1. Timeline for Marketing Strategy

Task	Start	End	Days
Market Analysis	1.01.2026	30.06.2026	180
Cost Analysis	1.10.2026	31.03.2027	181
Product Development	1.04.2027	31.03.2028	366
Prototype Release	1.10.2028	31.12.2028	92

- ***Market Analysis (1.01.2026-30.06.2026)***: This section will examine the target market, complete customer analysis, and make a technical comparison of competing products. Potential HVAC companies will be identified, and customer needs will be clarified by conducting a comprehensive market analysis.
- ***Cost Analysis (1.10.2026 – 31.03.2027)***: Production costs and B2B sales pricing will be determined during this period.
- ***Product Development (1.04.2027-31.03.2028)***: As technical improvements progress throughout the process, pilot products will be presented to potential partners, and pre-ordering processes will begin during this period.
- ***Prototype Launch (1.10.2028-31.12.2028)***: The launch of the developed water harvesting product, participation in trade fairs, signing of commercial contracts, and market entry will be carried out during this period.

Key Performance Indicators (KPI's)

The Key Performance Indicators (KPIs) determined to measure the project's performance are as follows:

- ***Financial Target***: Annual sales revenue of USD 211,267 and net profit of USD 47,393 will be achieved by the end of the 4th year of the project.
- ***Customer Productivity***: Licensing agreements will be signed with 20% of the targeted 10 HVAC manufacturers

- **Market Growth:** By supporting market expansion with innovative MOF technology, the project will contribute to a 10.4% growth rate in the atmospheric water harvesting sector. (shown in **Figure 3.1-1**)
- **Brand Awareness:** Increasing brand awareness through participation in trade fairs and technical conferences, aiming to be among the top 5 known domestic technologies.

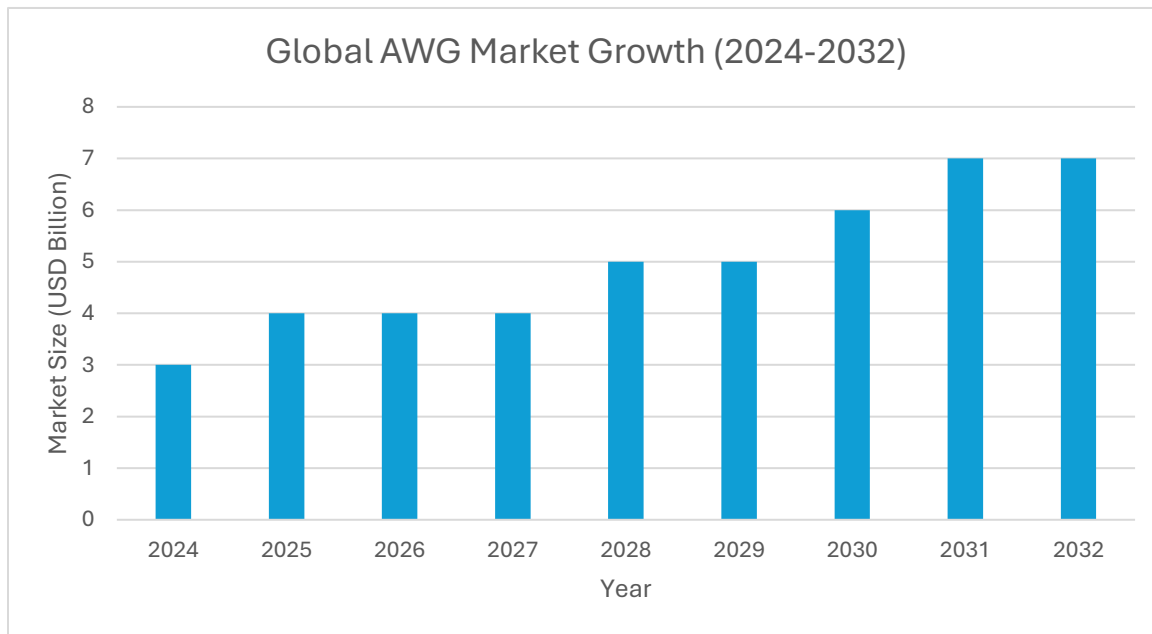


Figure 3.1-1. Global AWH Market Growth (2024-2032)

Source: Acumen Research and Consulting, 2024

Target Market and Customer Needs

The AquaCaptor project targets the air conditioning industry (HVAC), which has a very wide range of applications. Within this sector, it specifically targets the "room air conditioners" market. According to research, room air conditioners have a wide usage area of 42%. (See **Figure 3.1-2**)

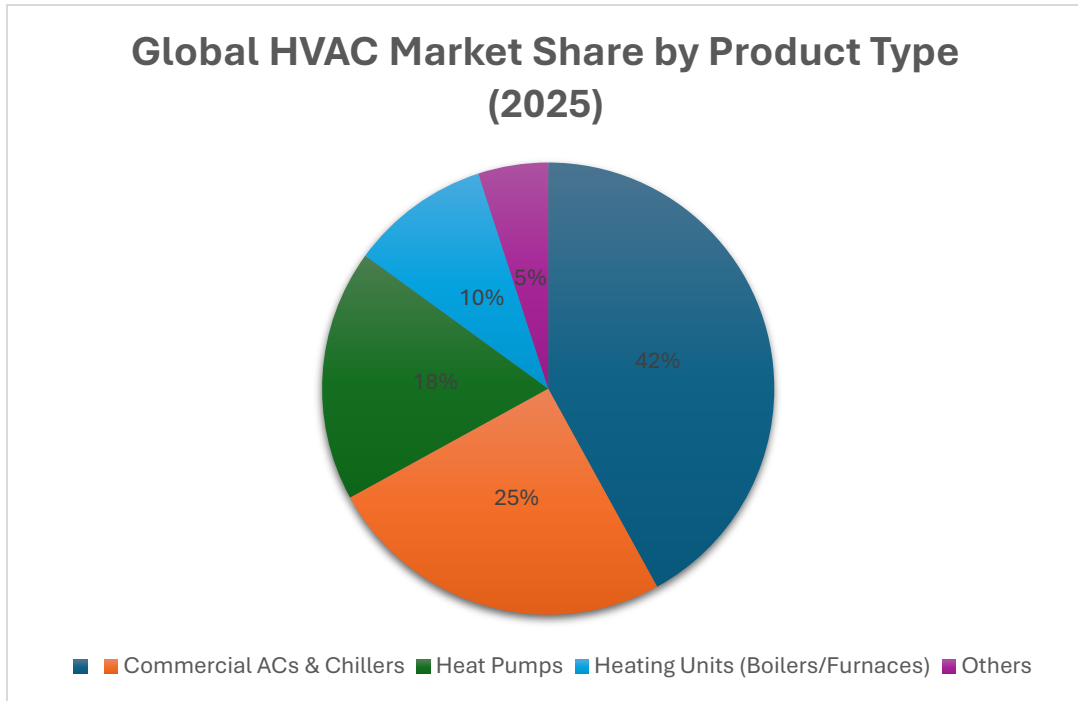


Figure 3.1-2. Global HVAC Market Share by Product (2025)

Source: Mordor Intelligence & IMARC Group, 2024

This system was chosen because of the high market demand for room air conditioners, compatibility with outdoor unit mounting, and high usage rates in the target sales region (Akdeniz region).

Since our target market model is B2B (Business to Business), our target customers are as follows:

- *Local Companies:* Arçelik, Vestel and Baymak
- *Global Companies:* Daikin, Mitsubishi Electric, and Samsung Climate Solutions.

Target Customer Needs

Customer needs analysis was conducted by considering our own customers (HVAC Manufacturers) and their target audience.

HVAC Manufacturers' Needs

- ***Need for Product Differentiation:*** Manufacturers need to sell more of their products than their competitors in the market.
- ***Compliance with Sustainability Regulations:*** Companies must comply with regulations to reduce their carbon footprint and ensure energy efficiency. To meet this need, they are in need of investing in sustainable projects.

Target Audience Needs

- ***Access to Water:*** End consumers living in humid and arid regions experience difficulties accessing water, and therefore their need for clean water is increasing due to water scarcity.
- ***Energy Cost Concerns:*** Consumers need economical ways to solve the problem of water access difficulties.

The AquaCaptor project meets the needs of both HVAC manufacturers and end consumers. By utilizing the waste heat from air conditioners to produce water, it ensures sustainability and meets water needs.

3.1.2. Sales Strategy

The sales plan involves partnering with companies that manufacture HVAC systems, rather than selling directly to end consumers. This allows the developed system to add value to the products of industry leaders.

Sales Targets

Based on cost analysis, the 4-year targets are as follows:

- ***2025:*** This is the project's starting year, and therefore no revenue target has been set.
- ***2026 - Pre-Sales & Grants:*** A revenue target of USD 67,605 is aimed for. This year's revenue mainly includes R&D grants.
- ***2027 - Growth:*** The aim is to increase revenue to USD 126,760. As product development matures, paid pilot projects and preliminary agreements are targeted.
- ***2028 - Launches:*** The target is to reach USD 211,267 in sales revenue. A net profit of USD 47,393 is targeted at the end of this year.

Table 3.1-1. Cost analysis.

YEAR	SALES REVENUE (USD)	OPEX (USD)	PBT (USD)	TAX (USD)	NET PROFIT (USD)	CUMULATIVE CF (USD)
2025	0	11,774	-11,745	0	-11,745	-11,745
2026	67,605	84,694	-17,090	0	-17,090	-28,835
2027	126,760	116,384	10,375	2,594	7,781	-21,054
2028	211,267	148,075	63,193	15,800	47,393	26,339

Sales Tactics

The tactics to be applied to achieve the goals are as follows:

- 1- ***Licensing:*** The goal is to license the rights of the developed original MOF/Salt composite formulation to large manufacturers. This will ensure a revenue stream.
- 2- ***Sales to Large Companies:*** The developed prototype meets the sustainability and energy needs of large companies, and partnership offers will be made to them.
- 3- ***Value-Oriented Sales:*** During the sales process, the environmentally friendly nature of the developed product will be emphasized in our advertisements and on our website.

Steps to Meet Targets

To achieve the defined sales tactics, the following steps will be followed in line with our sales calendar:

- ***Step 1 - Customer Diversity (Q1-Q2 of 2026):*** During market analysis, potential HVAC manufacturers will be categorized according to their capacities, sustainable approaches, and compatibility with the technology they produce.
- ***Step 2 - Pricing (Q4 of 2026 and Q1 of 2027):*** A B2B pricing model will be created by conducting a current competitor analysis and optimizing production costs.
- ***Step 3 - Strategic Partnerships (2027):*** Prototypes will be presented to the R&D teams of the targeted companies, and licensing negotiations will begin.
- ***Step 4 - Contracts (Q4 of 2028):*** As a result of all progress, sales targets will be achieved with the signing of long-term contracts upon the full completion of the prototype.

3.2. Operations

AquaCaptor is not planned for the establishment of a manufacturing plant and begin mass production; rather, it is conceived as an R&D initiative that develops, validates, and prototypes innovative technology. Our operational strategy is based on licensing the developed "MOF/Salt Composite" technology and supplying it to HVAC manufacturers. Therefore, our operational processes are limited to laboratory-scale synthesis, prototype assembly, and performance testing.

3.2.1. Technology

The technological basis of the project is based on the synthesis of a hybrid composite material (Chitosan/Alginate Coated $\text{CaCl}_2@ \text{UiO-66}$) that can capture atmospheric moisture even at low relative humidity levels. Our operational process is to synthesize this material and turn it into an "Add-on Kit" for air conditioning integration.

Prototype Production and Process Flow

Our prototype development process consists of the following basic steps:

1. **Green Synthesis:** The UiO-66 MOF structure is synthesized using environmentally friendly methods, without using toxic solvent (DMF) with high environmental impact.

Table 3.2-1. Technical properties of UiO-66 MOF.

BET Surface Area: 1296.8 m^2/g ^[1]

Pore Diameter: 21.48 Å ^[1]

Pore Volume: 0.54 cm^3/g ^[1]

Adsorption Capacity (water uptake): 0.25-0.35 g/g ^[2]

Thermal Stability: Up to 540°C ^[3]

Water Stability: 12 months^[3]

2. **Salt Loading (Confinement):** Calcium Chloride (CaCl_2) salt shows hygroscopic properties that allows us to achieve a high moisture capturing capacity. It is loaded into the synthesized MOF pores using a "wet impregnation" method to maximize water retention capacity.

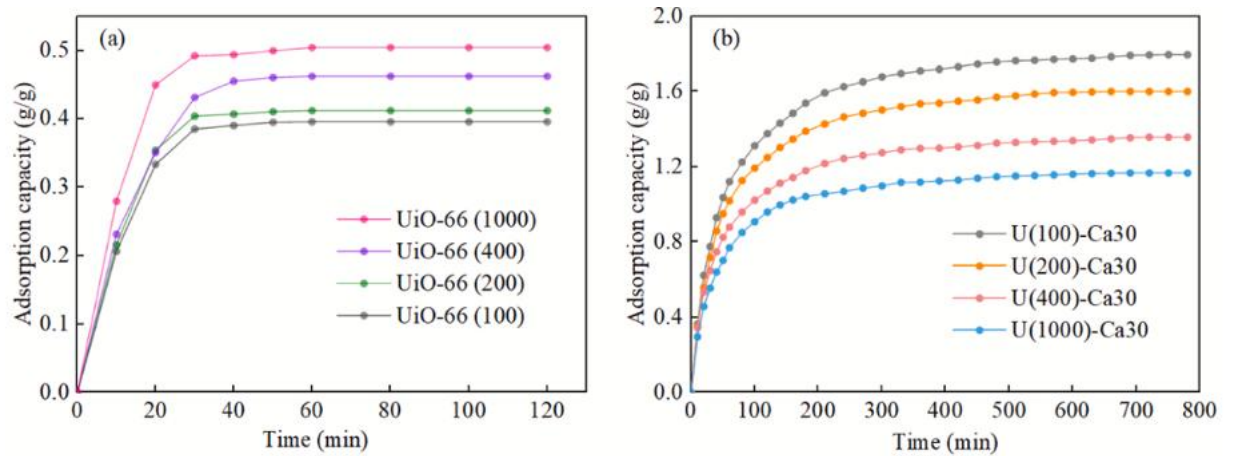


Figure 3.2-1. Adsorption capacity at 30°C and 80% RH,

a) UiO-66 of different crystal sizes, b) UiO-66- CaCl_2 composites of different matrix crystal sizes. ^[2]

3. **Dual Encapsulation:** To prevent salt leaching and ensure structural integrity, the material is first coated with alginate, then with chitosan polymers.

Figure 3.2-2. explains the formation of chitosan-alginate polyelectrolyte complex (PEC). Electrostatic interaction between the positively charged $-\text{NH}_3^+$ groups of chitosan and the negatively charged $-\text{COO}^-$ groups of alginate forms PEC ^[4].

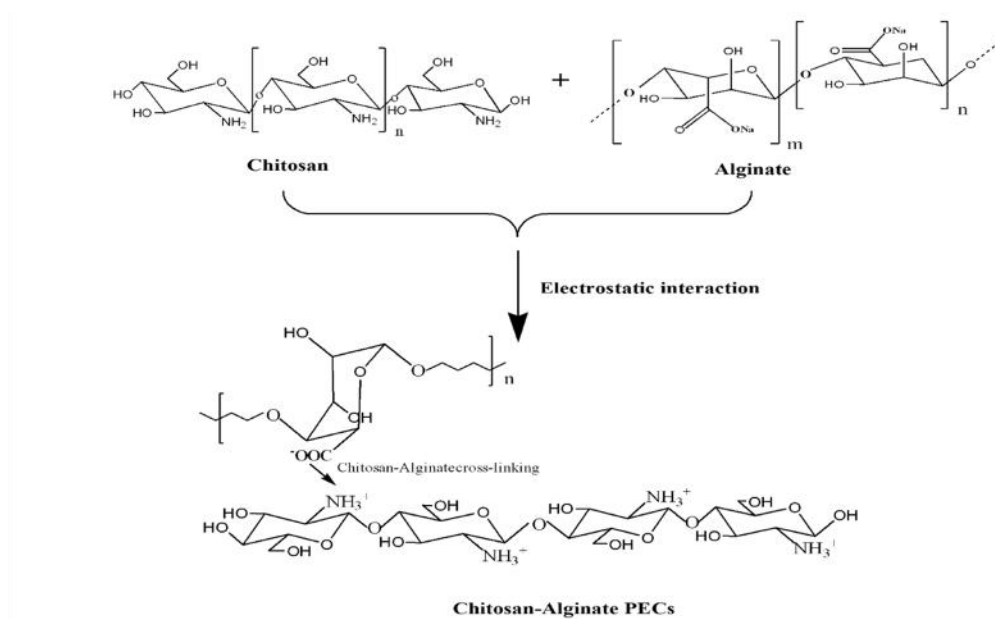


Figure 3.2-2. Cross-linking interactions between chitosan and alginate during PECs formation. ^[4]

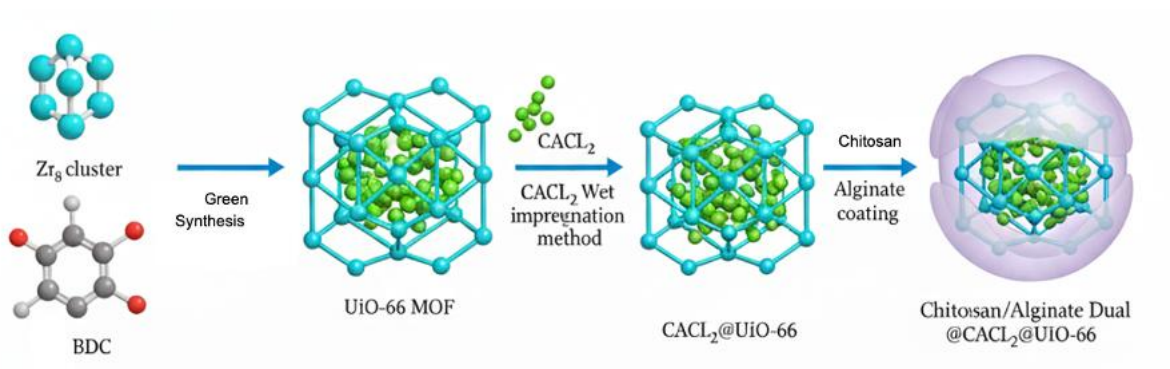


Figure 3.2-3. Schematic illustration of the synthesis process for the Chitosan/Alginate coated CaCl₂@UiO-66 composite (Designed by the Project Team).

The operational limits and key performance values defined for the system are presented in **Table 3.2-2**.

Table 3.2-2. Operational Environmental Parameters & Design Conditions.

Water Sorption Improvement: +20-40%
Mechanical Improvement: +30-70% Compressive Strength
Thermal Stability: Increased about 20-40°C
Optimum RH: 40-80%
Desorption Temperature: 50-80°C
Excess Heat: 40-90°C
Minimized Salt Leakage Risk^[5]
Improved Cycle Stability^[5]
Adaptable to Equipment Integration
Water Production/day: Depends on the RH% and adsorbent amount. Shown in Figure 3.2-3. and Figure 3.2.-4.

Table 3.2-3 details the water adsorption capacity of the synthesized CS/AL CaCl₂@UiO-66 composite under varying atmospheric conditions. These performance values were determined by the project team based on the specific isotherm characteristics of the dual-confined MOF structure.

Table 3.2-3. Water adsorption capacity of CS/AL CaCl₂@UiO-66 based on RH% level.

RH%	Adsorption Capacity (g H₂O/g adsorbent)
20	0.6
40	1.0
60	1.6
80	2.5

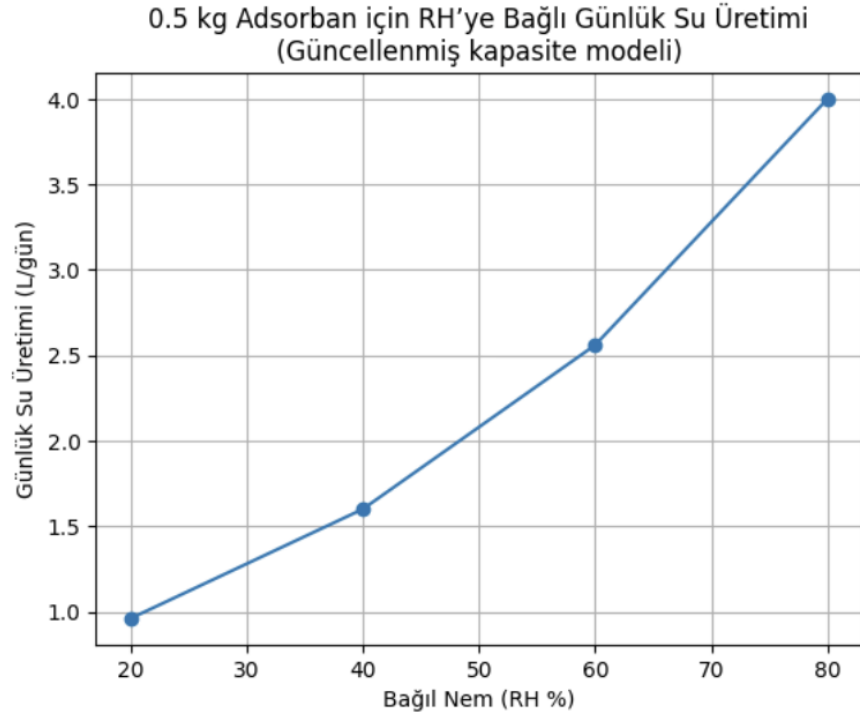


Figure 3.2-3. The estimated daily water production of CS/AL-CaCl₂@UiO-66 based on relative humidity (RH%).

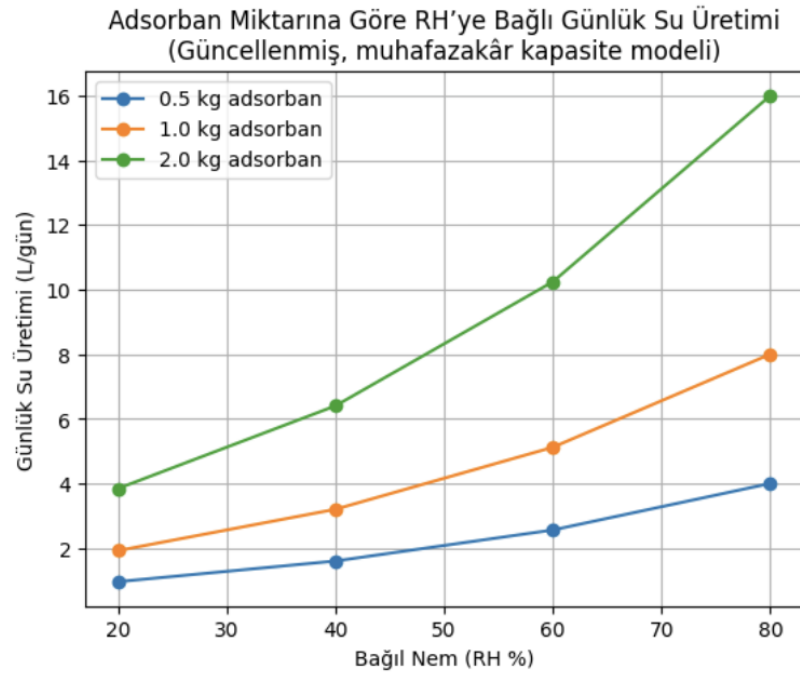


Figure 3.2-4. The estimated daily water production of different amounts of CS/AL-CaCl₂@UiO-66 based on relative humidity (RH%).

4. **Module Integration:** The resulting composite material is placed in filter cartridges produced with a 3D printer and mounted in the prototype housing along with a copper piping system to provide heat transfer.

In the sample representation in **Figure 3.2-5**, the composite material is stored in a breathable manner. This system is integrated into the device in a way that allows it to take excess heat from the engine.

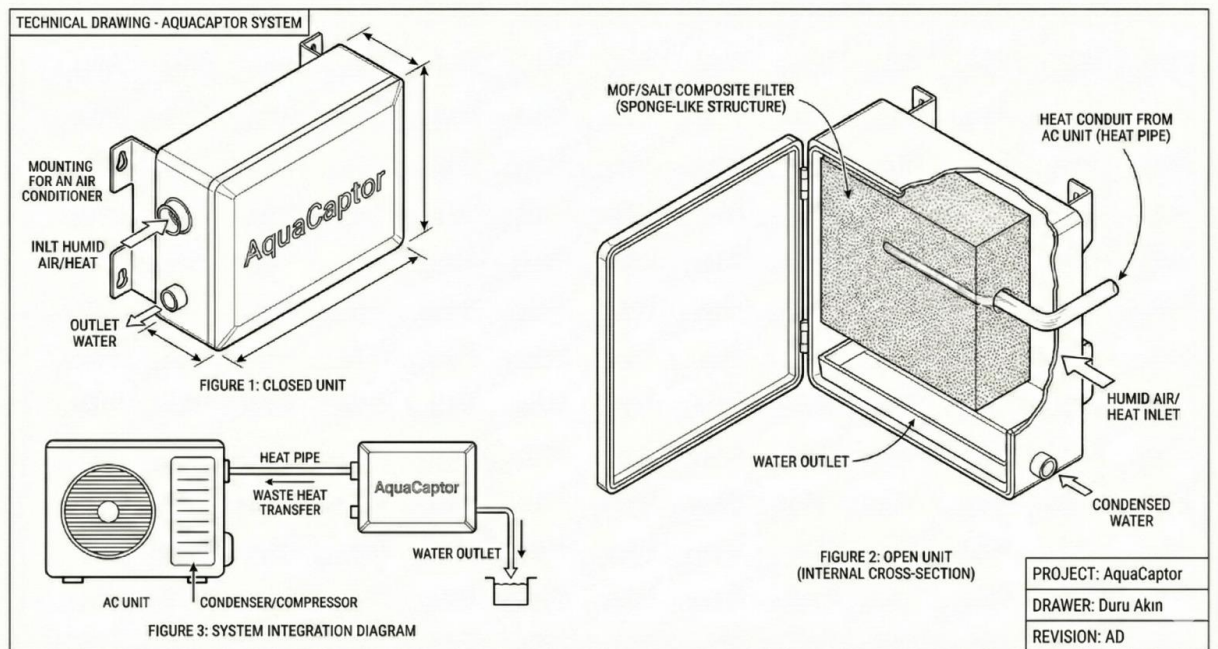


Figure 3.2-5. Technical Drawing of the estimated product.



Figure 3.2-6. Representation of the product.

Created by Gemini.

3.2.2. Locations & Facilities

Our company has implemented an "Asset-Light" model that does not require factory investment. All operations will be carried out using existing academic and technopark infrastructures within the framework of university-industry collaboration:

- **Synthesis and Laboratory Operations:** Hacettepe University Department of Chemical Engineering Laboratories will be used as the main center for the chemical synthesis and development of the material.
- **Prototyping and Assembly:** The "Prototyping Workshop" within Hacettepe Teknokent Workshop will be utilized for the design of the device's outer casing, 3D printing processes, and mechanical assembly.
- **Testing and Validation:** Services for advanced material characterization (SEM, XRD, BET) will be procured from external centers such as METU MERLAB and Bilkent UNAM.

3.2.3. Equipment & Tools

The equipment required for prototype production and R&D studies consists of precision laboratory instruments, not large industrial machines.

Table 3.2-1. Equipment and Tools required for Prototype Development

Adsorption Analyzer (BET)	Purpose of Use	Ownership / Access
High-Pressure Reactors (Auto-claves)	Solvothermal synthesis of UiO-66 MOF crystals.	University Lab
Magnetic Stirrers & Heaters	Preparation of homogeneous solutions for salt impregnation.	University Lab
Centrifuge & Vacuum Pumps	Separation and washing of synthesized MOF particles.	University Lab
Drying Ovens	Activation of adsorbent materials and drying of composite beads.	University Lab
3D Printer (FDM Type)	Rapid manufacturing of the device casing and cartridges.	Hacettepe Teknokent Workshop
CNC Machine	Fabrication of metal heat exchanger parts and connections.	Hacettepe Teknokent Workshop
XRD & SEM Devices	Crystallographic and morphological analysis of the material.	Outsourced (Service)
Adsorption Analyzer (BET)	Measuring the surface area and pore volume of the composite.	Outsourced (Service)

3.2.4. Environmental Aspects

Our project focuses on environmental sustainability, and environmental impacts will be carefully controlled during our laboratory operations. The environmental aspects of our project are listed below:

- ***Sustainable Design:*** The prototype is designed to use waste heat as an energy source, which brings the operational carbon footprint close to zero. In addition, the chitosan used is a biodegradable polymer and does not create a waste load.
- ***Waste Management (in Laboratory):*** Acidic solutions and washing water used in the synthesis phase will be separated and stored in accordance with the university's "Hazardous Waste Management Directive" and disposed of by licensed companies. PLA/ABS plastic waste generated during prototyping will be sent for recycling.
- ***Emission Control:*** Air quality will be protected by performing operations under a fume hood for the vapors of solvents used in MOF synthesis.

3.3. Metrics & Deliverables

The success of the project will be measured not only by the production of the final product, as well as by the attainment of the defined financial, technical, and operational objectives. This section defines the key metrics (KPIs) that will be used to monitor project performance and the defined project deliverables that will be produced throughout the project.

3.3.1. Key Metrics

Marketing and sales targets are detailed in Section 3.1.1. This section defines Complementary Metrics (Technical & Project KPIs) that will measure the project's technical success, operational efficiency, and management performance.

1. Technical Success Metrics (Technical KPIs):

- ***Water Production Efficiency:*** Achieving a minimum daily water production capacity of 5 liters under 60% relative humidity (RH) conditions with the developed prototype.

- *Specific Energy Consumption (SEC)*: Keeping the electrical energy consumed for water production (including fans and pumps) below 0.5 kWh/L.
- *Material Lifespan*: Ensuring the MOF composite operates for at least 1000 cycles with less than 5% capacity loss during regeneration.

2. Project Management Metrics:

- *Schedule Variance*: Completing critical path activities defined in the WBS with a maximum deviation of 5% from the planned dates.
- *Budget Compliance*: Keeping R&D and prototyping expenditures within the budget limits foreseen (10% tolerance).
- *Risk Mitigation Rate*: Preventing 90% of identified high-priority risks (Section 6) through proactive mitigation measures.

3. Quality and Safety Metrics:

- *Water Quality Index*: Ensuring the produced water has a TDS (Total Dissolved Solids) value of <100 ppm and a pH value in the range of 6.5-8.5.
- *Safety Incidents*: Maintaining a "Zero Work Accident" target during laboratory and workshop operations.

3 3.2. Deliverables

Technical reports, prototypes, and documentation to be produced throughout the project lifecycle are defined as "Deliverables". **Table 3.3** below lists the key deliverables, their dates, and recipients, planned in accordance with the WBS and Project Schedule.

Table 3.3. Deliverables.

Identification	Deliverable	Planned Date	Receiver
D1	Market Analysis Report (Detailed report containing market research, competitor analysis, and customer needs)	30.06.2026	Project Manager & Angel Investors
D2	Material Characterization Data (SEM, XRD, and BET analysis results of the synthesized MOF/Salt composite)	31.12.2026	R&D Lead & Academic Advisors
D3	Environmental Safety & Compliance Report (ISO 12100 and REACH compliance assessment)	31.12.2026	Quality Assurance Lead
D4	Cost Analysis & Business Model (Production costs, sales pricing, and financial projections)	31.03.2027	Project Manager & Potential Investors
D5	Prototype Design Package (Technical drawings, CAD models, and assembly schematics)	30.09.2027	Hacettepe Teknokent Workshop Team
D6	Final Working Prototype (Working AquaCaptor module integrated into the AC unit)	31.03.2028	Customers (HVAC Partners)
D7	Quality Control & Test Results (Performance tests, water quality analyses, and efficiency reports)	30.06.2028	Quality Assurance Lead
D8	Final Project Closeout Report (Summary of all processes, lessons learned, and financial close-out)	31.12.2028	TÜBİTAK & Angel Investors

3.4. Quality Assurance

In the AquaCaptor project, the Quality Assurance (QA) process is designed to verify the technical sufficiency of the developed prototype, prove water quality compliance with international standards, and minimize pre-commercialization risks. Our quality processes cover all stages from raw material intake (MOF synthesis) to final product output (harvested water).

Our quality assurance activities are grouped into three main categories: Material Verification, System Performance Tests, and Water Quality Analysis.

3.4.1. Material Verification

The chemical and physical stability of the developed MOF/Salt composite (Chitosan/Alginate coated $\text{CaCl}_2@ \text{UiO-66}$) is the fundamental factor determining the product lifespan. The following tests will be applied at this stage:

- ***Crystallographic and Structural Analysis:*** XRD (X-Ray Diffraction) analysis will be conducted to verify that the synthesized UiO-66 has the correct crystal structure.
- ***Surface Area and Porosity:*** The surface area, which determines the water-holding capacity of the material, will be measured via BET Analysis. The targeted surface area is a minimum of $1000 \text{ m}^2/\text{g}$.
- ***Morphological Control:*** The homogeneity of the Chitosan coating and the absence of cracks will be visualized using SEM (Scanning Electron Microscope).

3.4.2. Prototype Performance & Validation

The prototype will be tested in an environment simulating real operating conditions to provide "Proof of Concept."

- ***Adsorption/Desorption Kinetics:*** The speed at which the material adsorbs and releases water at different relative humidity rates (30% - 90% RH) will be tested.
- ***Cycle Stability:*** The material will be subjected to 1000 heating and cooling cycles to monitor the decline in water-holding capacity. The goal is to keep capacity loss under 5%.
- ***Leakage Tests:*** Salt leakage (liquefied salt dripping into the system), which is one of the most critical technical risks, will be tested. The Chitosan coating must prevent this leakage by 100%.
- ***Thermal Resistance:*** It will be verified that the composite material can withstand the $50\text{-}80^\circ\text{C}$ waste heat coming from the AC outdoor unit without degradation.

3.4.3. Water Quality & Safety

The ultimate success criterion of the project is that the produced water is of "potable" or "usable" quality.

- **Chemical and Biological Analysis:** Harvested water samples will be analyzed according to WHO (World Health Organization) drinking water guidelines.
- **Bacterial Growth Control:** It will be verified that bacterial formation in the water tank is prevented thanks to the antibacterial property of Chitosan.
- **Regulatory Compliance:** The product will be evaluated for compliance with CE Certification and ISO 12100 Machinery Safety regulations.

The following **Table 3.4** summarizes our Quality Control Plan mapped to the WBS packages.

Table 3.4. Quality Control Plan & Acceptance Criteria

Test ID (WBS Ref)	Test Description	Method / Standard	Acceptance Criteria
QC-01 (1.4.1)	Adsorbance Capacity	Gravimetric Analysis	> 2.0 g/g water uptake at 90% RH
QC-02 (1.4.6)	Salt Leakage Test	Visual Inspection & Conductivity Meter	0% Salt Leakage outside the capsule
QC-03 (1.4.3)	Product Efficiency	Input Heat vs. Water Output Ratio	Regeneration at 50°C waste heat
QC-04 (1.4.7)	Water Potability	WHO Guidelines for Drinking-water Quality	pH: 6.5-8.5, TDS < 500 ppm, No E. coli
QC-05 (1.4.4)	Stability / Durability	1000 Thermal Cycles	< 5% Capacity degradation
QC-06 (1.4.8)	Thermal Resistance	TGA Analysis	Stable structure up to 100°C
QC-07 (1.6.2)	CE Compliance	LVD & EMC Directives Check	Pass / Compliant

4. Organization

The project is executed by a multidisciplinary research team. The responsibility of the Project Leader is project coordination, planning and evaluation of results. The team has researchers for material synthesis, adsorption-desorption experiments and water harvesting system design.

Experiments are distributed for each team members including MOF-salt hybrid synthesis, characterization and performance testing in controlled humidity and controlled temperature conditions. To evaluate progress and discuss experimental results, the projects meetings are held with all team members.

Project Leader and Project Supervisor decide the strategic and technical steps to ensure scientific rigor and alignment with project objectives.

4.1. Organizational Boundaries and Interfaces

The project is embedded in the laboratory of Hacettepe University. It includes the technologies of materials science, thermodynamic, and water harvesting. The project is maintained with laboratories, procurement, and infrastructure management.

The primary external stakeholders of the project are TÜBİTAK and the Hacettepe Teknokent. TÜBİTAK helps the project improvement with providing financial support and evaluating project progress and outcomes. The project must be dependent on TÜBİTAK regulations, milestone reporting and performance criteria. The Teknokent gives the opportunity of interacting with innovative-oriented environments and technology-focused companies. The project outcomes are expected to get support for prototype development or startup formation.

The parent organization is Hacettepe University and It holds the primary administrative and managerial responsibility of project. Hacettepe University gives us the research environment that is needed. Financial management, laboratory safety, and procedures are conducted with the parent organization's regulations. The project team operates technical decisions.

The customer organization is TÜBİTAK that provide financial support and evaluation. Its role is limited to defining funding conditions, monitoring progress and evaluating predefined milestones.

4.1.1. Resource Owners

Hacettepe University is the owner of the laboratory, climate chambers, and measurement equipment for the experiments.

4.1.2. Receivers

TÜBİTAK is primary receiver of the project. TÜBİTAK evaluate the results with periodic technical and financial reports. Hacettepe University and Chemical Engineering department receive scientific outputs, research activities and potential technology development.

4.1.3. Sub-contractors

Material synthesis, experiments, water harvesting measurements are made by team members so there are no Sub-contractors.

4.1.4. Suppliers

External suppliers provide products for synthesis and experiments of projects. Supplied items are chemicals for production of MOF, salt and humidity control components.

Table 4.1-1 Company suppliers and their deliveries.

Company/ Contact	Deliverable	Comment
Merck (Sigma-Aldrich)	Zirconium salts (e.g. ZrCl_4)	Used for UiO-66 synthesis
Merck (Sigma-Aldrich)	Calcium chloride (CaCl_2)	Hygroscopic salt used in hybrid formulation
TCI Chemicals	Chitosan and sodium alginate	Biopolymers used for coating
Avantor	Laboratory consumables	Used for synthesis, washing, and processing steps
Memmert	Humidity control equipment	Used for environmental control and performance testing

4.1.5. Cross Functions

Table 4.1-2 Cross functions involved in the project.

Function	Dept.: Contact	Responsibility
<i>Product Management</i>	Project Team / Project Leader	Determination of project scope and performance target of the Chitosan/Alginate coated $\text{CaCl}_2@\text{UiO-66}$ hybrid in air-conditioning systems
<i>Marketing</i>	Project Team / Technopark Interface	Identification of potential HVAC application areas
<i>Sales</i>	Technopark Interface / Technology Transfer Office	Evaluation of potential customers such as HVAC manufacturers
<i>Service</i>	Engineering Support	Maintenance, and integration requirements of the product in air-conditioning systems
<i>Training</i>	Project Team / Research Team	Internal training on laboratory safety and experimental procedures
<i>Manufacturing</i>	Project Team / Research Team	Synthesis and processing of the Chitosan/Alginate coated $\text{CaCl}_2@\text{UiO-66}$ hybrid material
<i>Quality</i>	Research Team / Project Supervisor	Verification of experimental repeatability, performance efficiency, and validation of results
<i>Technology</i>	Research Team	Development system design
<i>Supply Management</i>	University Procurement Unit	Procurement of chemicals, consumables, and equipment

4.2. Project Organization

The structured project organization model is utilized. Project Leader is responsible for project planning, coordination, and achievement of goals. Also, the project leader communicates with Hacettepe University, TÜBİTAK, and the Technopark interface.

The project is divided into different functional areas. These areas are synthesis, coating, adsorption-desorption performance testing, and efficiency for air-conditioning integration. Each area is assigned to project team members.

Scheduling, documentation, progress monitoring, and reporting are responsibility of Project Leader. Execution function is made by a research team in Hacettepe University Chemical Engineering Laboratory.

Communication in the project follows a hierarchical structure. Technical information is shared with meetings and documentation. Authority, responsibility, and reporting lines are clearly defined to ensure efficient coordination and timely decision-making.

4.2.1. Project Team

Table 4.2-1 Project managers.

Role	Organization: Name
Project Manager	Hacettepe University – AquaCaptor Team Ceren Oktay
Technical Project Coordinator	Hacettepe University – AquaCaptor Team Duru Akin
Project Supervisor	Hacettepe University – AquaCaptor Team Emine Naz Özeş

Table 4.2-2 Project internal functions.

Function	Organization: Name	Comment
R&D and Material Development	Hacettepe University – Chemical Engineering	MOF Synthesis, salt impregnation, polymer coating
Prototype Design & Assembly	Hacettepe Technopark Workshop	Mechanical design, 3D printing, module integration
Quality Assurance & Testing	Hacettepe University / External Labs	Performance tests, water quality, durability
Market & Cost Analysis	Hacettepe University – AquaCaptor Team	B2B market research, pricing, financial feasibility
Project Management & Reporting	Hacettepe University – AquaCaptor Team	Schedule tracking, risk management, reporting

Table 4.2-3 Team members involved in the project.

Organization: Name	Availability (%)	Comment
Ceren Oktay	100	Project management, process design, reporting
Beyza Bıçak	100	Material synthesis and characterization
Duru Akın	100	Prototype development and testing
Emine Naz Özeş	100	Market analysis and financial modeling
Caner Kovancı	100	Quality control, validation and risk analysis

4.2.2. Communication and Reporting

Table 4.2-4 Communication and reporting methods.

Type of Communication	Method / Tool	Frequency / Schedule	Information	Participants / Responsibilities
Internal Communication:				
<i>Project Meetings</i>	In-person / Tele-conference	Bi-weekly	Project status, experimental progress, technical issues, task planning	Project Manager, Technical Project Manager, Project Team
<i>Sharing of project data</i>	Digital	Continuously	Experimental data, reports	Project Manager, Project Team Members
<i>Milestone Meetings</i>	In-person / Tele-conference	At defined milestones	Milestone achievement	Project Manager, Technical Project Manager, Project Supervisor
<i>Final Project Meeting</i>	In-person / Tele-conference	End of project	Project summary, results	Project Manager, Project Team, Project Supervisor
External Communication and Reporting:				
<i>Project Progress Reports</i>	TÜBİTAK reporting system	According to TÜBİTAK schedule	Project status, progress	Project Manager, TÜBİTAK
<i>Financial Reports</i>	TÜBİTAK reporting system	According to TÜBİTAK schedule	Budget usage	Project Manager, Parent University Administration, TÜBİTAK
<i>Stakeholder Meetings</i>	Teleconference / Meeting	At major milestones	High-level project status	

Communication and reporting ensure the project information can be followed by team members and stakeholders in a timely. Project team meetings are held on a bi-weekly. In these meetings, there must be Project Manager, Technical Project Manager and research team members. The results and upcoming tasks should be discussed in meetings.

External communication and reporting are managed by the Project Manager. TÜBİTAK receives technical and financial reports periodically. These reports consist of progress summaries, results, and milestone achievements. Communication with the Technopark is conducted at predefined intervals for technology transfer and air-conditioner integration opportunities. Meetings with external stakeholders are held at major milestones.

5. Schedule and Budget

5.1. Work Breakdown Structure

The steps planned for this project are listed using a work breakdown structure (WBS). The work breakdown structure decomposes this project into manageable tasks to accurately define the scope of the project and to provide efficient, planned management. WBS was given in **Appendix C: Work Breakdown Structure for AquaCaptor Project**.

Work packages are as follows:

1.1. Project Management

This section establishes the foundation of the project. This reflects the execution of market aware R&D project with steps taken to minimize risks. Involves decision making processes, technical and commercial activities.

1.2. Material Selection

This section defines the core of the project. This examines the optimal material selection to provide best performance, ensuring quality, safety and durability. Each material undergoes individual evaluation before production.

Adsorbent was selected based on its water capturing capacity and low toxicity. MOF was selected based on its thermal and water stability, large surface area, and relatively low environmental impact. Materials for coating were selected that would reduce the risk of salt leakage, improve mechanical properties and stability and enhance performance. Then the production was designed by assembling all the components.

This work package eliminates the failure risk of the material.

1.3. Product Development

This section is the core innovation phase. Provides a functional composite and AWH prototype. Designed production of the material is performed and structural integrity is maintained. Prototype production combines laboratory scale innovation and application in realistic conditions.

1.4. Quality Control Tests

This section provides performance feedback of the produced material. Adsorption capacity, stability, leakage, water quality, temperature resistance, water harvesting performance, environmental impacts and mechanical durability are tested. This work package reduces technical and operational risks by controlling important mechanisms. Also, collects scientific data for environmental regulations and market acceptance.

1.5. Final Evaluation of Tests

This section provides optimization using data obtained with quality control tests. Through all the experimental results, performance limiting factors can be identified and leads to improvement of the system. With all the optimizations, AquaCaptor presents a product that has no risk of failure and delivers high performance.

1.6. Environmental Safety

This section ensures the system's safety and sustainability. Before the product is launched on the market, it must be proven to comply with certain regulations. Compliance for Water Efficiency Regulations, CE Certification, ISO:12100 Machinery Safety, REACH Regulation must be completed.

1.7. Market Analysis

Market and competitor analysis is a critical step for a company before launching a new product. Market analysis creates a clear target. Developing a marketing strategy provides advantages to product and the company. This work package minimizes economic and commercial risks and ensures that precautions are taken.

1.8. Cost Analysis

This work package provides extensive economic assessment. An analysis performed of equipment, material, production, test, labor costs. This step is necessary for market sales and investors. Financial viability is determined and a competitive product is presented. Profit optimization should be carried out through cost and income analysis.

5.2. Schedule and Milestones

A Gantt Chart is created to present a project schedule. The timeline for the tasks in the work packages created in the WBS is determined in the Gantt Chart. It shows durations and start-end dates for each task. This chart also shows the workload and overlapping activities, ensuring efficient use of resources. The Gantt chart is given in **Appendix D: Project Schedule (Gantt Chart) for AquaCaptor Project**.

For this project, total project timeline is set as 434 days, between October 23,2025 and December 31, 2028. The time allocated for each work package is as follows:

- Project Management: 159 days.
- Market Analysis: 180 days.
- Material Selection: 274 days.
- Environmental Safety: 183 days.
- Cost Analysis: 181 days.
- Product Development: 366 days.
- Quality Control Tests: 182 days.
- Final Evaluation of Tests: 183 days.
- Prototype Release: 92 days.

Table 5.2-1. Milestones of AquaCaptor Project.

Milestones	Description	Milestone Criteria	Planned Date
M0	Project Start	Project goals and scope are defined. Timeline for the project is determined.	23.10.2025
M1	Completion of Project Planning	Resources and budget are determined. Market analysis and literature review are completed.	31.03.2026
M2	Completion of Material Selection	Composite materials and structure, and production strategy are defined.	31.12.2026
M3	Completion of Product Development	Successful transformation of selected products into the functional product.	31.03.2028
M4	Completion of Final Evaluation of Tests	All experimental data is analyzed and optimized products meet performance and safety criteria. Ready for product release.	30.09.2028

5.3. Budget

As a high-tech R&D project in Türkiye, most of the early-stage capital is non-dilutive funding from **TÜBİTAK 1512 (BİGG)** or **1507 (SME R&D)**.

The economic feasibility analysis of the AquaCaptor project was conducted in accordance with comprehensive chemical engineering design principles and engineering economy methodologies. The Total Capital Investment (TCI) and operating costs were estimated using the factorial estimation methods outlined by *Peters et al.* [6], *Towler & Sinnott* [7], and *Turton et al.* [8]. Profitability indicators, including Net Present Value (NPV) and Internal Rate of Return (IRR), were calculated based on the standard procedures described by *Blank & Tarquin* [9], *Sullivan et al.* [10], and *Newnan et al.* [11]. Furthermore, the cost-benefit analysis framework was aligned with the guidelines provided by *Boardman et al.* [12] and the *European Commission* [13]. Energy efficiency evaluations utilized the metrics from the *IEA* [14] and *Short et al. (NREL)* [15]. Corporate tax and funding calculations comply with the current regulations of the *Republic of Türkiye Ministry of Treasury and Finance* [16] and *TÜBİTAK BİGG principles* [17].

Stakeholder: Republic of Türkiye / TÜBİTAK.

Funding Equation (G):

$$G = (C_{Personnel} \times 0.75) + (C_{Equipment} \times 0.75)$$
$$G = \$200,025$$

This covers 75% of R&D labor, material costs.

Project's Equity (E):

$$E = P - G_{Offset} = 9,350\$ \text{ (Founders' Cash Outlay)}$$

Fixed Capital Investment (FCI):

$$FCI = (C_{Hardware} + C_{IP} + C_{Setup}) \times (1 + f_{inst})$$

C_{Hardware}: \$6,500 (3D Printers, Vacuum Oven, Precision Scales)

C_{IP}: \$2,000 (Patents & Licensing Fees)

C_{setup}: \$1,500 (Lab Prep at Hacettepe)

f_{inst}: 0.10 (Installation factor for wiring/assembly)

$$FCI = (6,500 + 2,000 + 1,500) \times (1 + 0.10)$$

$$FCI = \$11,000$$

The Present Value Formula

$$P_n = F_n \times (P / F, i, n) = \frac{F_n}{(1 + i)^n}$$

Where:

- F_n = Net Cash Flow (Net Profit) in year n.
- $(1 + i)^{-n}$ = The discount factor for year n at 12% interest.

A. Year 0 (2025)

Since this is the base year, money is already at its present value ($n = 0$).

Net Cash Flow (F_0) = -\$135,275

Calculation: $-135,275 \times (1.12)^0 = -135,275 \times 1.0$

PV₂₀₂₅ = -\$135,275

B. Year 1 (2026)

Net Cash Flow (F_1) = -\$84,695

Calculation: $-84,695 \times (1.12)^{-1} = -84,695 \times 0.8929$

$$PV_{2026} = -\$75,624$$

C. Year 2 (2027)

$$\text{Net Cash Flow (F}_2\text{)} = -\$4,624$$

$$\text{Calculation: } -4,624 \times (1.12)^{-2} = -4,624 \times 0.7972$$

$$PV_{2027} = -\$3,686$$

D. Year 3 (2028)

$$\text{Net Cash Flow (F}_3\text{)} = +\$63,192$$

$$\text{Calculation: } +63,192 \times (1.12)^{-3} = 63,192 \times 0.7118$$

$$PV_{2028} = +\$44,980$$

Salvage Value

$$\text{Salvage Value (L)} = +\$5,000$$

$$\text{Calculation: } +5,000 \times (1.12)^{-3} = 5,000 \times 0.7118$$

$$PV_{\text{salvage}} = +\$3,559$$

Table 5.3-1. Discounted Cash Flow Analysis and Net Present Value (NPV) Calculation.

Year (n)	Net Cash Flow (F _n)	Discount Factor (1.12) ⁻ⁿ	Present Value (P _n)
2025	-\$135,275	1.0000	-\$135,275
2026	-\$84,695	0.8929	-\$75,624
2027	-\$4,624	0.7972	-\$3,686
2028	+\$63,192	0.7118	+\$44,980
Salvage	+\$5,000	0.7118	+\$3,559
PROJECT NPV			-\$166,046

The project's raw NPV of -\$166,046 indicates that the internal operations alone do not cover the high R&D costs within this 3-year window. However, the TÜBİTAK Grant (approx. ~+\$198,825) turns the Adjusted NPV to +\$32,779, confirming economic viability for the project.

Working Capital (WC): Calculated to cover 25% of the first year's operating expenses to ensure liquidity of cash flow.

$$WC = 0.25 \times OPEX_{2026} = 0.25 \times \$152,300 = \$38,075$$

Total Capital Investment (P) with Contingency (f_{con}):

$$P = (FCI + WC) \times (1 + f_{con})$$

$$P = (11,000 + 38,075) \times (1 + 0.15) = \$56,436$$

Companies like **Arçelik, Vestel, or Daikin** provide "Developmental Payments" or "Signing Fees" to secure the rights to our waste-heat recovery technology.

Stakeholder: Industry Partners (Licensees).

Based on the market penetration strategy of reaching **20% of the target HVAC manufacturers**, the revenue scales as follows:

A. Year 1 (2026): Pilot Phase

Equation: R₂₀₂₆ = Initial Partnership Fee

Calculation: Total Revenue = \$67,605

B. Year 2 (2027): Market Entry (32 Units)

Equation: $R_{2027} = 32 \times \$2,112.67 + \$59,154$ (Licensing Fee)

Calculation: Total Revenue = \$126,760

C. Year 3 (2028): Scaling Phase (100 Units)

Equation: $R_{2028} = 100 \times \$2,112.67$

Calculation: Total Revenue = \$211,267

Table 5.3-2. Projected Revenue Stream and Sales Forecast (2025-2028).

Year	Units Licensed	Revenue Type	Total Revenue (USD)
2025	0	R&D Phase	\$0
2026	0	Pilot/Partnership	\$67,605
2027	32	Licensing + Royalties	\$126,760
2028	100	Full Royalties	\$211,267

Detailed Operating Expenses (OPEX – Annual A)

Table 5.3-3. Detailed Annual Operating Expenses (OPEX) Breakdown.

Expenses	2025	2026	2027	2028
<i>Labor</i>	\$118,800	\$118,800	\$80,000	\$80,000
<i>Materials</i>	\$4,700	\$17,500	\$32,500	\$44,500
<i>Infrastructure</i>	\$4,500	\$4,500	\$4,500	\$4,500
<i>Admin</i>	\$7,275	\$11,500	\$14,384	\$19,075
OPEX	\$135,275	\$152,300	\$131,384	\$148,075

Note: While the cost estimation methodology follows the academic standards cited above, specific unit prices for raw materials (e.g., Zirconium salts, polymers) and equipment are based on 2025 average market prices obtained from commercial suppliers and global databases.

Personnel costs are estimated based on current industry salary standards for R&D engineers in Turkey. All financial projectors assume a strict B2B business model.

OPEX is the summation of materials (C_M), labor (C_L), infrastructure (C_{inf}) and admin (C_A).

$$O_n = C_{Labor} + C_{Materials} + C_{inf} + C_A$$

A. Year 0 (2025): Startup Costs

Labor: 5 people x \$23,760 (Avg. Salary) = \$118,800

Materials: 50 kg MOF x \$80/kg + Consumables = \$4,700

Infrastructure (C_{inf}): Energy + Maintenance = \$4,500

Admin: Legal + Initial Marketing = \$7,275

$$O_{2025} = 118,800 + 4,700 + 4,500 + 7,275 = \mathbf{\$135,275}$$

B. Year 1 (2026)

Labor: 5 people x \$23,760 (Avg. Salary) = \$118,800

Materials: 50 kg MOF x \$80/kg + Consumables = \$17,500

Infrastructure (C_{inf}): Energy + Maintenance = \$4,500

Admin: Legal + Marketing = \$11,500

$$O_{2026} = 118,800 + 17,500 + 4,500 + 11,500 = \mathbf{\$152,300}$$

C. Year 2 (2027)

Labor: 5 people x \$23,760 (Avg. Salary) = \$118,800

Materials: 50 kg MOF x \$80/kg + Consumables = \$32,500

Infrastructure (C_{inf}): Energy + Maintenance = \$4,500

Admin: Legal + Marketing = \$14,384

$$O_{2027} = 118,800 + 32,500 + 4,500 + 14,384 = \mathbf{\$131,384}$$

D. Year 3 (2028)

Labor: 5 people x \$23,760 (Avg. Salary) = \$118,800

Materials: 50 kg MOF x \$80/kg + Consumables = \$44,500

Infrastructure (C_{inf}): Energy + Maintenance = \$4,500

Admin: Legal + Marketing = \$19,075

$$O_{2028} = 118,800 + 44,500 + 4,500 + 19,075 = \mathbf{\$148,07}$$

Cost of Capital (CoC) Analysis

The Cost of Capital represents the Minimum Attractive Rate of Return (MARR). In this project, the CoC is calculated as a Weighted Average Cost of Capital (WACC), reflecting the mixture of TÜBİTAK grant funding (0% interest) and Group 10 Equity (investor expectation).

$$WACC = (w_g \times r_g) + (w_e \times r_e)$$

- **Grant Weight (w_g):** 75% (TÜBİTAK 1512/1507 support ratio).
- **Grant Cost (r_g):** 0% (As it is a non-dilutive public grant).
- **Equity Weight (w_e):** 25% (Internal capital contribution).

- **Equity Expectation (r_e):** 12% (Standard MARR/rate for high-tech R&D).

$$WACC = (0.75 \times 0.00) + (0.25 \times 0.12) = 3.0\%$$

12% MARR Expected. (Minimum Acceptable Rate of Return)

$$r_e = R_f + \beta \times ERP$$

- **Risk-Free Rate (R_f):** 15.00% (Based on 10-year Government Bond yields in Türkiye).
- **Equity Risk Premium (ERP):** 5.50% (Standard market premium for equity investment).
- **Project Specific Risk (β):** 1.15 (Beta coefficient reflecting higher volatility in HVAC/R&D tech).

$$r_e = 15.00 + 1.15 \times 5.50 = 21.325\% + 0.325\% \text{ (for Liquidity/Small-Cap Premium)}$$

$$\text{CoC} = 21.65\%$$

Internal Rate of Return (IRR) Analysis

$$0 = -P + \sum_{n=1}^N \frac{ATCF_n}{(1 + IRR)^n} + \frac{L}{(1 + IRR)^N}$$

Year 0 (P_{Adj}): -\$13,527 (Founders' remaining equity after 90% of Year 0 costs are covered by the first year).

Year 1 (F_1): -\$12,300 (Net burn rate after grant support).

Year 2 (F_2): -\$4,624 (Market entry operational cost).

Year 3 ($F_3 + L$): +\$68,192 = (Scaling profit \$63,192 + Salvage Value \$5,000).

For $i = 25\%$:

$$NPW = -13,527 - \frac{12,300}{1.25} - \frac{4,624}{1.25^2} + \frac{68,192}{1.25^3} = +8,605$$

For i = 30%:

$$NPW = -13,527 - \frac{12,300}{1.30} - \frac{4,624}{1.30^2} + \frac{68,192}{1.30^3} = +5,315$$

For i = 40%:

$$NPW = -13,527 - \frac{12,300}{1.40} - \frac{4,624}{1.40^2} + \frac{68,192}{1.40^3} = -354$$

By interpolation,

$$IRR = 30 + (40 - 30) \times \frac{5,315 - 0}{5,315 - (-354)} = 39.37$$

Calculated IRR for AquaCaptor: **39.37%**

39.37% > 21.65% IRR is higher than CoC, therefore the project is efficient, highly feasible.

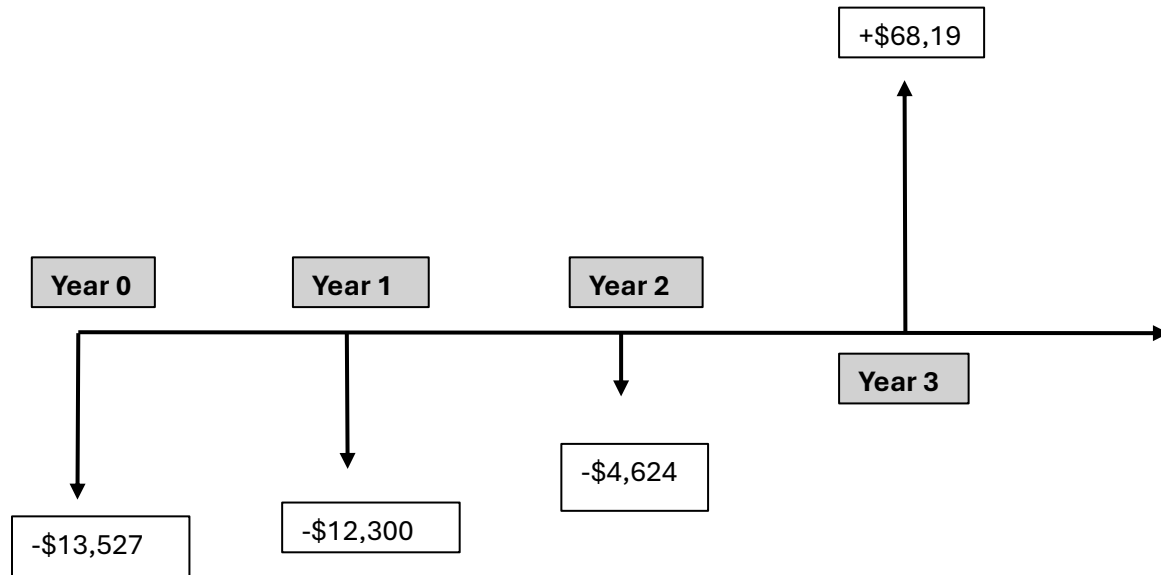
$$P = (FCI + WC) \times (1 + f_{con})P = (11,000 + 38,075) \times (1.15)$$

Final Budget (P): 56,436

Table 5.3-4. Total Capital Investment (TCI) and Final Project Budget.

Budget Category	Value (USD)	Source / Formula
<i>Fixed Capital (FCI)</i>	\$11,000	$(C_{\text{Hardware}} + C_{\text{IP}} + C_{\text{S}}) \times 1.10$
<i>Working Capital (WC)</i>	\$38,075	$\text{OPEX}_{2026} \times 0.25$
<i>Contingency (15%)</i>	\$7,361	$(FCI + WC) \times 0.15$
TOTAL BUDGET (P)	\$56,436	Total Startup Capital

Cash Flow of Each Years within Assumed Project Timeline



6. Risk Management

The project's comprehensive risk assessment, conducted by the Project Manager, is presented in **Table 6.1**. All potential threats listed in this table are prioritized based on their impact levels and directly linked to the corresponding WBS packages.

Following this assessment, the specific mitigation strategies and contingency scenarios designed to address each identified risk are detailed in **Table 6.2**.

Risk controls and monitoring will be regularly reviewed in weekly meetings, and each possibility for each risk will be assessed by the Project Manager. Work will be carried out on risks with a high probability of occurrence, and efforts to reduce this probability will be supported by our team. Finally, the current status of key issues and risks will be recorded in monthly Project Monitoring Reports to facilitate tracking.

Table 6.1 Table of Risk Assessment

				Risk Impact Level(1-5)					
Risk ID	Related WBS Item	Risk Description	Impact Description	Risk Category	Cost	Sch.	Qual.	Risk Probability Level (1-5)	Priority Level (=Impact*Probability)
AQC-R-01	1.3.3.	CaCl ₂ Leaching	Permanent capacity loss & corrosion risk	Technical	3	4	5	4	20
AQC-R-02	1.3.5.	HVAC Corrosion	Salt damaging HVAC metal pipes/parts	Technical	4	3	5	5	25
AQC-R-03	1.3.1.	UiO-66 Defect	Low surface area; poor salt loading	Technical	3	4	5	3	15
AQC-R-04	1.4.6.	Low Regen. Heat	HVAC waste heat insufficient for desorption	Technical	2	4	5	4	20
AQC-R-05	1.3.5.	Airflow Blockage	Filter increasing HVAC pressure drop.	Technical	1	3	4	4	16
AQC-R-06	1.5.3.	Material Costs	High Zirconium prices exceeding budget	Financial	5	2	3	4	20
AQC-R-07	1.3.2.	Hydrogel Swelling	Chitosan clogging air channels excessively.	Technical	2	3	4	3	12
AQC-R-08	1.4.2.	Bacterial Growth	Contamination in water due to Chitosan.	Quality	2	2	5	3	15
AQC-R-09	1.1.6.	Lab Equipment Busy	Delays in XRD/SEM characterization.	Schedule	1	5	1	3	15
AQC-R-10	1.3.4.	Inhomogeneous Mix	Uneven salt distribution in MOF	Technical	2	3	4	3	12
AQC-R-11	1.6.4.	VOC Emission	Release of Ethanol/Acetone vapors	Environmental	2	3	4	2	12
AQC-R-12	1.4.3.	Cycle Stability	Performance drop after many cycles	Quality	2	2	4	3	12
AQC-R-13	1.7.2.	Competitors	Cheaper silica-gel products in market	Market	4	1	1	2	8
AQC-R-14	1.5.1.	Reporting Delay	Delay in compiling test data for weekly reviews.	Schedule	1	2	1	2	4

Table 6.2. Mitigations and Contingencies

Risk ID	Related WBS Item	Risk Description	Mitigation Strategies	Contingency Actions
AQC-R-01	1.3.3.	CaCl ₂ Leaching	Application of the double encapsulation (UiO-66 + Chitosan) method.	Replacing the coating material
AQC-R-02	1.3.5.	HVAC Corrosion	Coating the outside of the equipment with corrosion-resistant material.	Replacing the affected part.
AQC-R-03	1.3.1.	UiO-66 Defect	Modifying parameters during microwave synthesis	Synthesis from scratch
AQC-R-04	1.4.6.	Low Regen. Heat	Optimizing the MOF according to the HVAC system.	Providing extra energy to the system.
AQC-R-05	1.3.5.	Airflow Blockage	MOF particle size readjustment	Adding a bypass line to the system
AQC-R-06	1.5.3.	Material Costs	Bulk purchase of zirconium	Exploring alternative raw material suppliers
AQC-R-07	1.3.2.	Hydrogel Swelling	Optimization of Chitosan/Alginate Bonds	Thinning the hydrogel layer
AQC-R-08	1.4.2.	Bacterial Growth	Ensuring the water tank is kept closed and cool.	Adding a sterilization unit to the water tank
AQC-R-09	1.1.6.	Lab Equipment Busy	Reservations must be made months in advance.	Using multiple analysis labs
AQC-R-10	1.3.4.	Inhomogeneous Mix	Use of ultrasonics	Extending the mixing time
AQC-R-11	1.6.4.	VOC Emission	Ensuring the use of filters.	Increasing ventilation systems
AQC-R-12	1.4.3.	Cycle Stability	Optimizations are implemented to ensure stability.	Trying different parameter values
AQC-R-13	1.7.2.	Competitors	Emphasizing green production and energy efficiency.	Changing the marketing strategy
AQC-R-14	1.5.1.	Reporting Delay	Set an early deadline and ensure continuous system monitoring.	Giving an oral presentation / company visit

APPENDIX

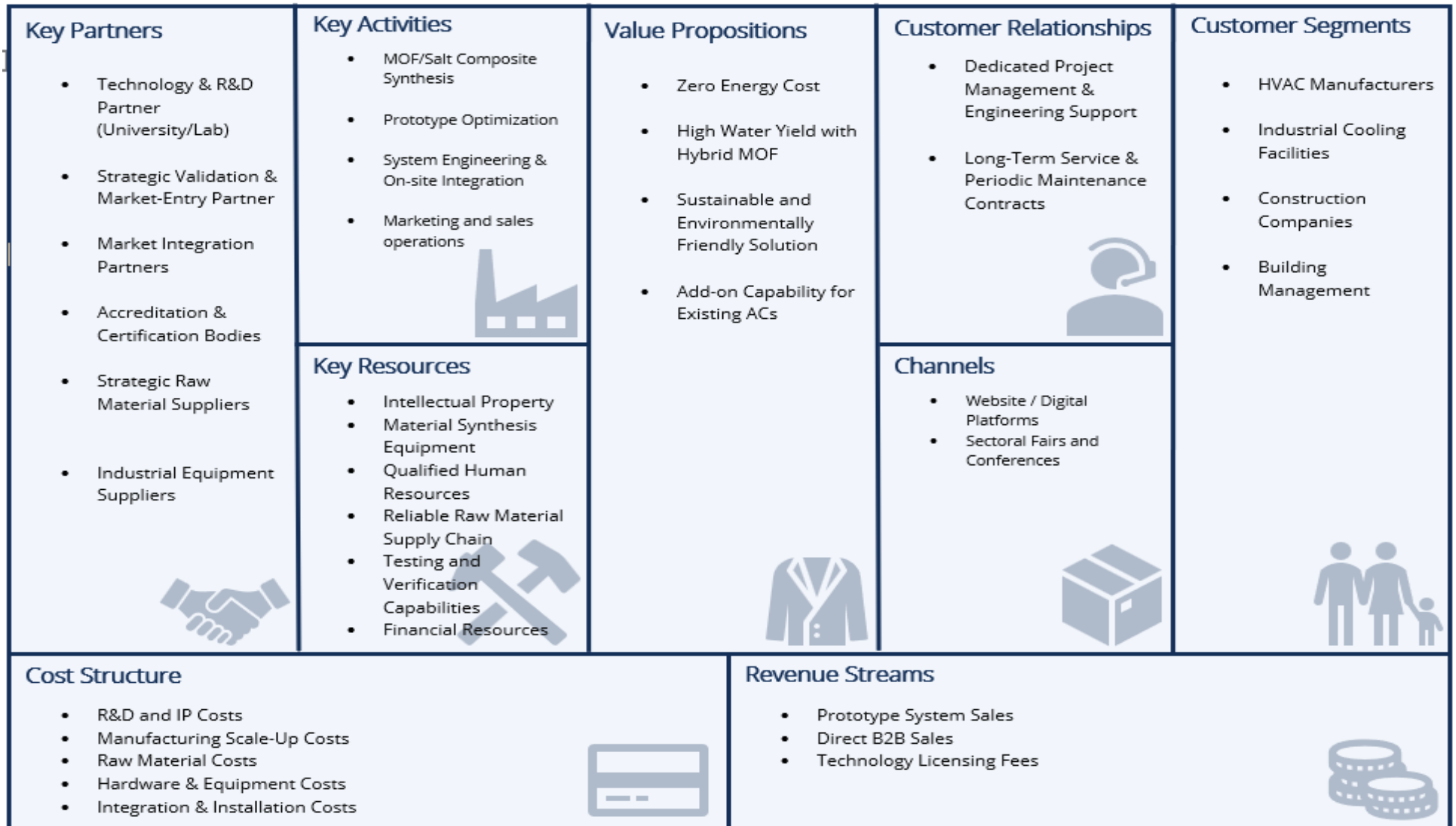
Appendix A: Project Charter.

Project Charter				
Project Title	Aqua Captor- Water Harvesting with MOF/Salt Composite		Project Manager Ceren Oktay	
Project Start Date	Oct 23, 2025	Project End Date	Dec 31, 2028	
Business Need				
Access to clean water is one of the most pressing problems today. The AquaCaptor project utilizes waste heat from existing HVAC systems to produce water by capturing moisture from the atmosphere. The project aims to offer an environmentally friendly and sustainable solution for water production by developing innovative MOF/salt composite technology.				
Project Scope		Deliverables		
The project involves synthesizing a CaCl ₂ @UiO-66 composite to capture atmospheric moisture and produce clean water. It also includes manufacturing a prototype of this material that utilizes waste heat from air conditioners.		Market analysis report, cost analysis and business model, materials report, prototype design and report, and quality control test results.		
Risk and Issues		Assumptions/Dependencies		
Low humidity levels lead to decreased performance, increased risk of corrosion in the air conditioning system, and high MOF costs.		Assumptions: Prevention of salt leachate, adequacy of waste heat Dependencies: Raw material supply time, External laboratory services		
Financials				
This Project's budget is 56434 USD.	Key Metrics			
	Metric	Baseline	Target	Unit
	Water Yield	0.35	≥5	L/day (@ 60% RH)
	Regeneration Energy	2.0	≤ 0.5	kWh/L
	Water Quality (TDS)	500	≤ 100	mg/L
	Prototype Cost	—	≤ 28864	USD
Milestone Schedules				
Milestone	Target Completion Date			
-Project Planning and Literature Review	March 31, 2026			
-Market Analysis& Research	June 30, 2026			
-Material Selection & Safety	Dec 31, 2026			
-Cost Analysis & Pricing	March 31, 2027			
-Prototype Development	March 31, 2028			
- Quality Tests	June 30, 2028			
- Launch & Final Release	Dec 31,2028			
Project Team		Approval/Review Committee		
Project Manager Ceren Oktay		Sponsor		
Beyza Bıçak		Business Division Head		
Team Members	Cenker Kovancı Duru Akın	Business Unit Head		
	Emine Naz Özeş	Finance Manager		

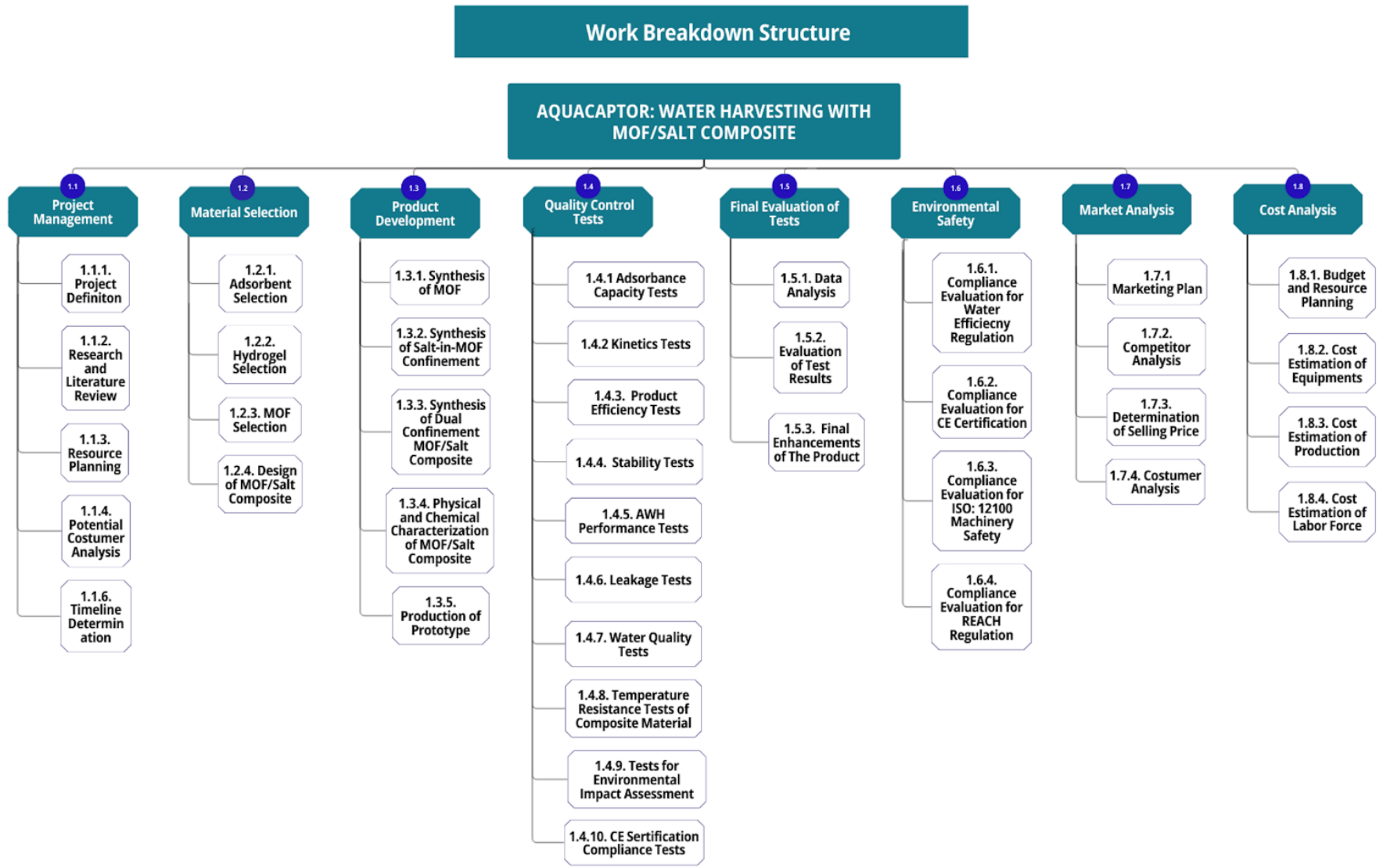
Appendix B: Business Canvas Model.

The Model Business Canvas

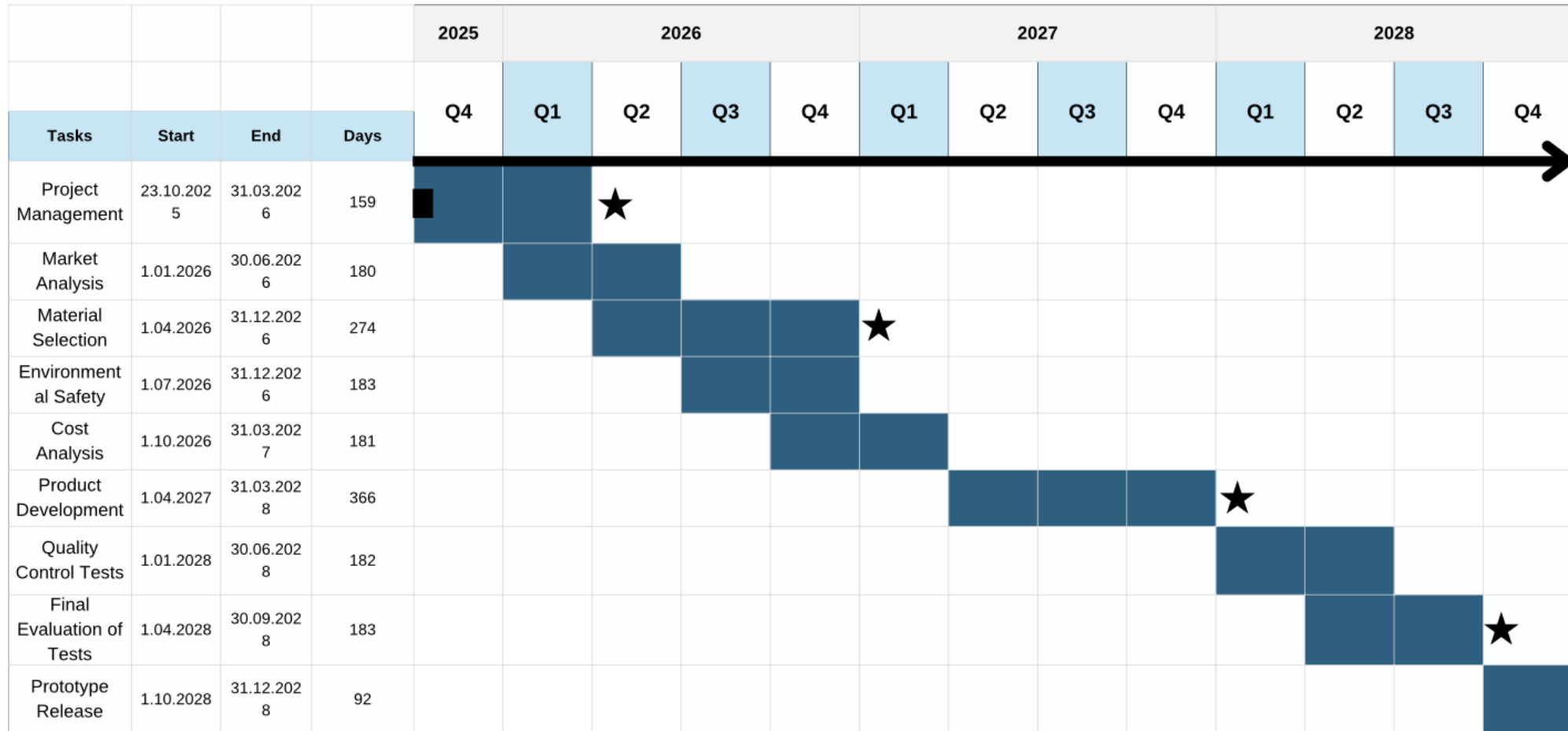
Atmospheric Water Harvesting with MOF/Salt Composite



Appendix C: Work Breakdown Structure for AquaCaptor Project.



Appendix D: Project Schedule (Gantt Chart) for the AquaCaptor Project.



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